

1. () 有關於小孩的 brain stem astrocytom 之描述，下列何者為誤？

- A) 好發於 pons.
- B) 可能長出腦幹外面圍繞著基底動脈
- C) 病灶的邊界是 ill-defined.
- D) 病灶都是 solid type.

(1 答案): **D (神經系統)**

(1 難易度): 易

(1 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p56.

(1 註解): P56, 可以有 cyst or calcification component but rare

2. () 下列有關成人神經膠質瘤 (adult glioma)的治療何者正確？

- A) WHO grade 1 的神經膠質瘤，不需手術切除，經切片證實後可以考慮觀察或在有神經學症狀下接受放射治療。
- B) WHO grade 3 的 IDH mutant, 1p19q codeleted oligodendroglioma 盡可能手術切除合併術後放射治療及輔助性化療 PCV (procarbazine/lomustine/vincristine)。
- C) IDH mutant astrocytoma WHO grade 2 可考慮術前化療 temozolomiden (TMZ)再手術即可，不須放射治療。
- D) Glioblastoma 目前並無治療指引以臨床試驗為主。

(2 答案): **B (神經系統)**

(2 難易度): 中等

(2 參考文獻): NCCN Guidelines Version 1.2022 CNS cancer.

(2 註解): GLIO-2: IDH mutant, 1p19q codeleted oligodendroglioma, WHO grade 3 Standard RT and neoadjuvant or adjuvant PCV (procarbazine/lomustine/vincristine) after surgery (category 1)

3. () 下列關於神經膠質母細胞瘤(glioblastoma)的研究，何者為誤？

- A) MGMT methylated 神經膠質母細胞瘤在 the EORTC/NCIC trial 中接受同步放化療具較長的存活期中位數
- B) 在 NEJM 2009 發表的文章中發現 IDH mutation 使神經膠質母細胞瘤具較好預後
- C) 神經膠質母細胞瘤術後接受同步放化療 temozolomiden (TMZ)比單獨輔助性放射治療，有較長 3 年整體存活率
- D) 神經膠質母細胞瘤術後接受同步放化療 temozolomiden (TMZ)療合併輔助性 TMZ+ 腫瘤治療電場 (Tumor-Treating Field; TTF) 持續治療與同步放化療 temozolomiden (TMZ) 合併輔助性 TMZ 存活期相當

(3 答案): **D (神經系統)**

(3 難易度): 難

(3 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p60.

(3 註解): P60, Maintenance TTF (Stupp JAMA 2015):...PFS (7.1 vs. 4.0 mo, $p = 0.001$) and MS (20.5 vs. 15.6 mo, $p = 0.004$) favored addition of TTF.

4. () 下列何者不是高惡性度神經膠質瘤(High grade gloma)的相關分子預後因子？

- A) TNF
- B) ATRX loss
- C) IDH status
- D) 1p/19q codeletion

(4 答案): A (神經系統)

(4 難易度): 中等

(4 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p53.

(4 註解): P53, Molecular prognostic factors (favorable): IDH1/2 mutation, 1p/19q codeletion, ATRX loss, TP53 wt., TERT promoter wt., MGMT promoter hypermethylation.

5. () 關於高惡性度神經膠質瘤(High grade gloma)描述，下列何者有誤？

- A) 成人最常見的原發惡性腦瘤
- B) 頭痛為臨床最常見的症狀
- C) 疾病預後與性別有關
- D) 好發年齡介於 45-55 歲之間

(5 答案): C (神經系統)

(5 難易度): 易

(5 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p52.

(5 註解): P52, Clinical prognostic factors: age, histology, KPS, extent of surgery, duration of symptoms.

6. () 聽神經瘤(vestibular schwannoma)最好發於下列那一條顱神經？

- A) 顏面神經(facial nerve)
- B) 耳蝸神經(cochlear nerve)
- C) 前庭神經(vestibular nerve)
- D) 中間神經(nerve intermedius)

(6 答案): C (神經系統)

(6 難易度): 易

(6 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, Halperin, E., Wazer, D., Pérez, C., Brady, L. 7th edition; Chapter 38. P 2611

(6 註解): P2611-2612, Vestibular schwannomas (VSs) or acoustic neuromas is a histologically benign schwann-cell sheath tumor that usually arises from the inferior division of the vestibular nerve (not the cochlear portion). VSs arise as a result of the loss of a tumor-suppressor gene on the long arm of chromosome 22 (in sporadic cases this is a somatic mutation; in neurofibromatosis Type 2 (NF2) this is either inherited or represents a new mutation that may then be transmitted to spring) .

7. () 有關腦室膜瘤(Ependymoma)之描述，下列何者為誤？

- A) 顱內好發於第 4 腦室
- B) 第 2 常見的原發性兒童腦瘤
- C) 常與神經纖維瘤第 1 型(NF1)合併發生於脊髓腫瘤
- D) 疾病預後主要與手術切除腫瘤程度及年紀有關

(7 答案): **C (神經系統)**

(7 難易度): 中等

(7 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p63.

(7 註解): P63, Extent of resection and age are important prognostic factors.

8. () 下列腦膜瘤(meningiomas)中，何者非由 Internal carotid artery supply？

- A) Falx cerebri meningioma
- B) Cranial convexity meningioma
- C) Olfactory groove meningioma
- D) Inner sphenoid wing meningioma

(8 答案): **B (神經系統)**

(8 難易度): 難

(8 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, Halperin, E., Wazer, D., Pérez, C., Brady, L. 7th edition; Chapter 38, P2534. Table 38.3

(8 註解): Falx & olfactory: low frontal median meningioma feed from ICA; Sphenoid wing meningioma: inner third with feeding from ICA.

9. () 引起腦膜瘤(meningiomas)復發的最重要原因是下列何者？

- A) 病人年齡及性別
- B) 顱骨被侵蝕與否
- C) 術後殘留腫瘤量
- D) 良性腦膜瘤的病理組織

(9 答案): **C (神經系統)**

(9 難易度): 中等

(9 參考文獻): Handbook of Evidence-based Radiation Oncology, Eric K. Hansen, Mack Roach III; Chapter 2, p82.

(9 註解):

P82, **The extent of surgical tumor removal is the most important factor in the prevention of recurrence.** An often overlooked aspect of the Simpson grading system is that it refers exclusively to the removal of intradural tumors, and thus leaving the tumor e.g. in the sagittal sinus could still be compatible with complete removal. Recurrence after gross total tumor removal occurred in 11–15% of cases, but was 29% when removal was incomplete (length of follow-up not specified)⁸; 5-year recurrence rates of 37%³⁸–85%³⁹ after partial resection are also quoted. The overall recurrence rate at 20 years was 19% in one series,⁴⁰ and 50% in another. Malignant meningiomas have a higher recurrence rate than benign ones.

10. () 下列何種腦瘤最易經腦脊髓液轉移?

- A) 神經管胚細胞瘤(medulloblastoma)
- B) 室管膜瘤(ependymoma)
- C) 星狀細胞瘤(astrocytoma)
- D) 乳頭狀瘤(papilloma)

(10 答案): A (神經系統)

(10 難易度): 中等

(10 參考文獻): Perez and Brady's Principles and Practice of Radiation Oncology, Halperin, E., Wazer, D., Pérez, C., Brady, L. 7th edition; Chapter 38. P2594

(10 註解): P2594-2595, Medulloblastomas : Seeding and metastases ≈10–35% have seeded the craniospinal axis at the time of diagnosis.